



CSE2200: DESIGN PROJECT - II

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1. Use case Diagram



Figure 1: Use Case Diagram of AgroInvest System

The Use Case Diagram illustrates the functional interactions between the users and the AgroInvest system. The system consists of four primary actors: Farmer, Investor, Admin, and SSLCommerz Payment Gateway. Each actor performs specific tasks according to their role within the platform.

The Farmer can register or log in to the system, create new farming projects, upload verification documents (such as National ID, land ownership documents, and land photos), update project milestones or progress, and view the profit report after project completion. During project creation, uploading verification documents is a mandatory step, which is represented using the `<<include>>` relationship.

The Investor can browse available agricultural projects, view project details, invest in a selected project, monitor investment information, view profit reports, and rate farmers after project

completion. The investment process requires payment processing through the integrated SSLCommerz Payment Gateway, which is associated with the Process Payment use case. Additionally, the View Profit Report and Rate Farmer use cases extend the View My Investment functionality, indicating that these are optional actions available after an investment has been completed.

The Admin is responsible for managing and monitoring the overall system. The administrator can log in to the system, verify farmer documents, approve or reject submitted projects, manage farmer and investor accounts, monitor investments and projects, enter the final project profit or loss, and generate system reports. Project approval includes farmer document verification, ensuring that only verified farmers and approved projects become available for investment.

Overall, the Use Case Diagram provides a high-level representation of the AgroInvest platform by defining the responsibilities of each actor and illustrating how they interact with the system to support secure agricultural crowdfunding, project management, investment processing, and profit distribution.

2. Activity Diagram

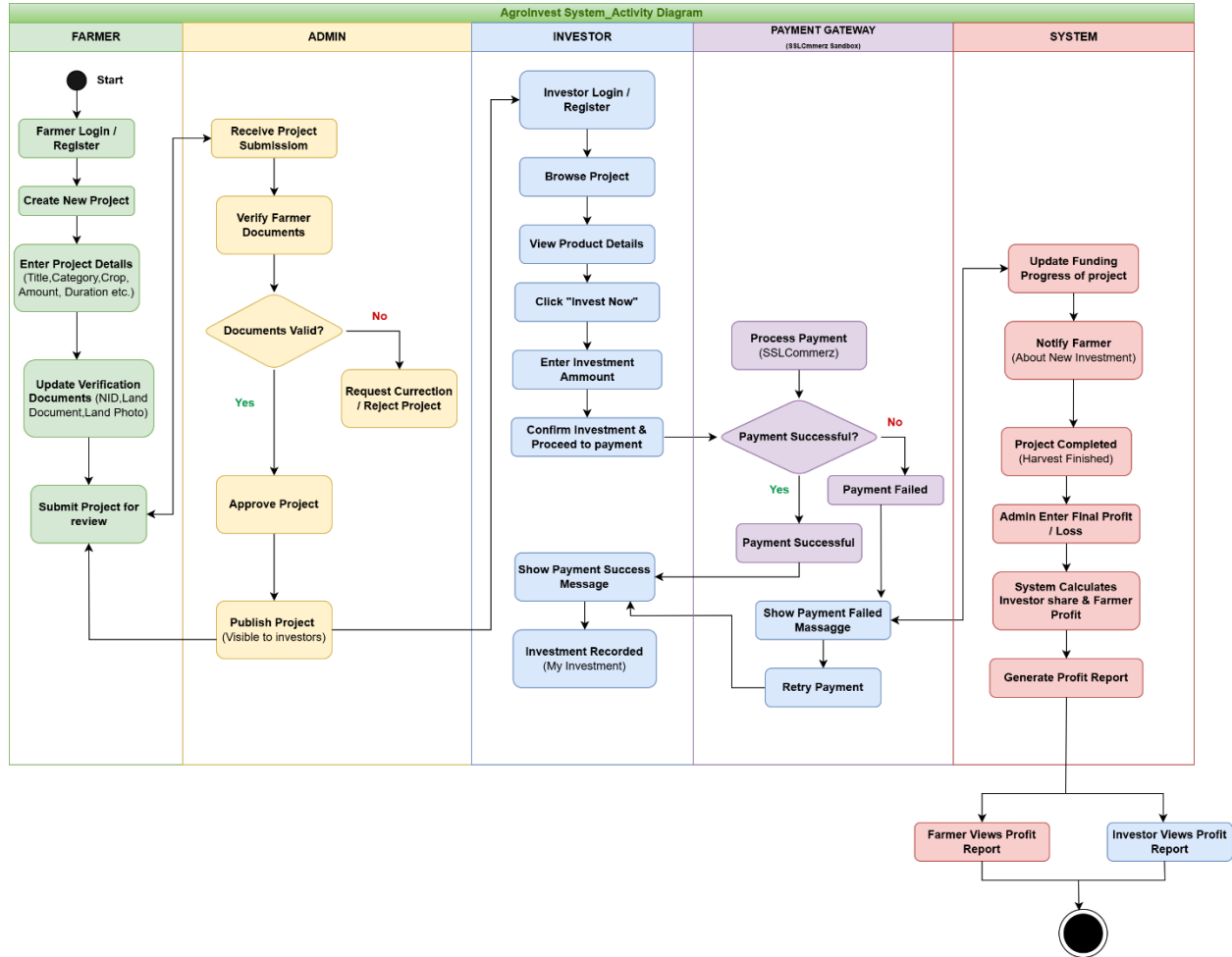


Figure 2: Activity Diagram of AgroInvest System

The Activity Diagram illustrates the overall workflow of the AgroInvest system by showing how different participants—Farmer, Admin, Investor, SSLCommerz Payment Gateway, and the System—interact throughout the agricultural investment process. The diagram represents the complete business process, starting from project creation and ending with profit report generation.

The workflow begins with the Farmer, who registers or logs into the system and creates a new farming project by entering project information such as the project title, category, crop type, funding amount, and estimated duration. The farmer then uploads the required verification documents, including the National ID, land ownership documents, and land photographs, before submitting the project for administrative review.

Once the project is submitted, the Admin receives the project request and verifies the farmer's documents. A decision is made based on the validity of the submitted documents. If the documents are invalid or incomplete, the administrator requests corrections or rejects the project, requiring the farmer to resubmit the necessary information. If the documents are verified successfully, the administrator approves the project and publishes it, making it visible to investors on the platform.

After project approval, the Investor logs into the system, browses the available farming projects, views detailed project information, and selects a project to invest in. The investor enters the investment amount, confirms the investment, and proceeds to payment through the integrated SSLCommerz Payment Gateway. The payment gateway processes the transaction and determines whether the payment is successful or unsuccessful. If the payment fails, the investor is notified and given the option to retry the payment. If the payment is successful, the system records the investment and displays a confirmation message.

Following a successful investment, the System automatically updates the project's funding progress and notifies the farmer about the new investment. Once the farming project is completed and the harvest is finished, the administrator enters the final project profit or loss. Based on the predefined profit-sharing policy, the system calculates the profit distribution for both the farmer and the investors and generates the final profit report. Finally, both the farmer and the investor can access and view their respective profit reports, completing the overall business process of the AgroInvest platform.

3. ER Diagram

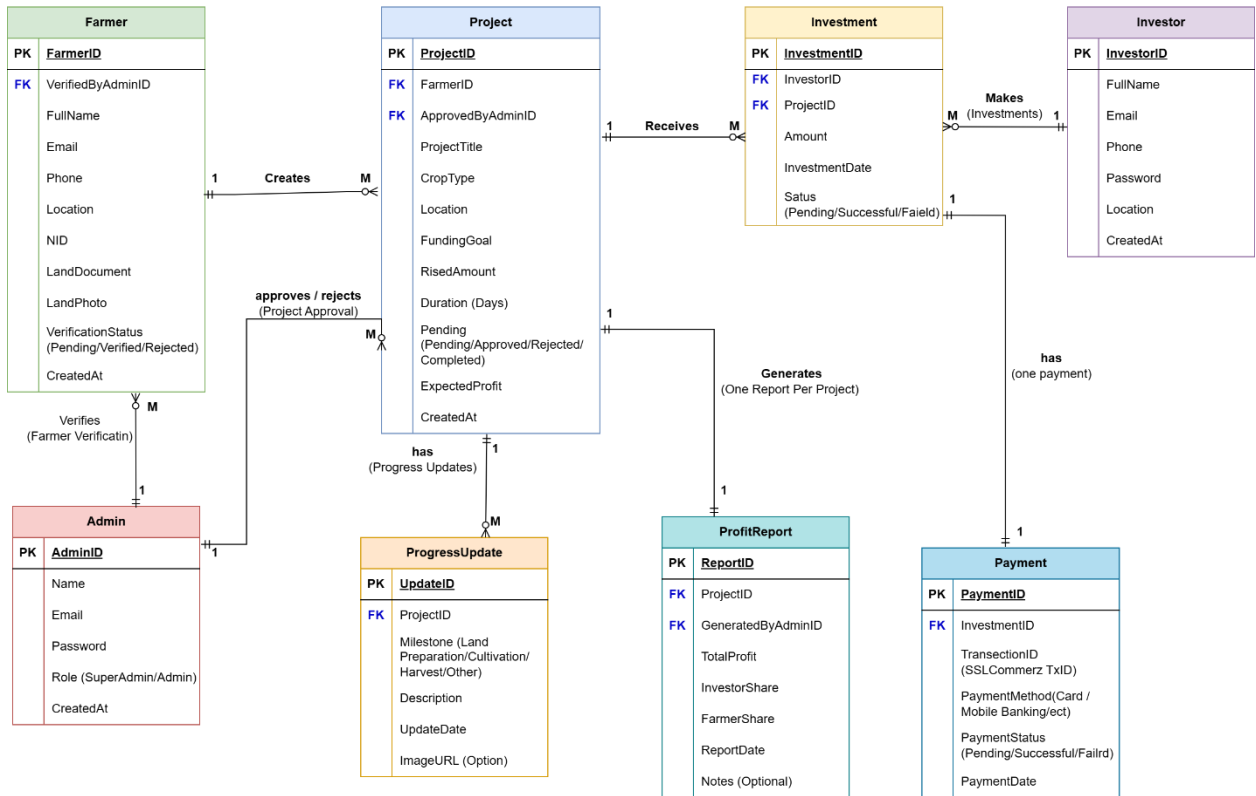


Figure 3: Entity Relationship (ER) Diagram of AgroInvest System

The Entity Relationship (ER) Diagram illustrates the database structure of the AgroInvest system by defining the main entities, their attributes, primary keys, foreign keys, and the relationships among them. The diagram represents how data is stored, connected, and managed to support the overall functionality of the agricultural crowdfunding platform.

The system consists of eight major entities: Farmer, Admin, Project, Investor, Investment, Payment, ProgressUpdate, and ProfitReport. Each entity stores specific information required for managing agricultural projects, investments, payments, project progress, and profit distribution.

The Farmer entity stores farmer information, including personal details, verification documents, and verification status. A farmer can create multiple projects, resulting in a one-to-many relationship between the Farmer and Project entities. Every project belongs to only one farmer.

The Admin entity manages system administration activities such as verifying farmers, approving or rejecting submitted projects, and generating profit reports. One administrator can verify multiple farmers and approve multiple projects. In addition, each profit report is generated by an administrator after the completion of a project.

The Project entity contains all project-related information, including project title, crop type, funding goal, raised amount, duration, status, and expected profit. Each project can receive investments from multiple investors and can also have multiple progress updates throughout its lifecycle. After project completion, each project generates a single profit report, representing a one-to-one relationship between the Project and ProfitReport entities.

The Investor entity stores investor account information. An investor can invest in multiple projects through the Investment entity, while a project can receive investments from multiple investors. Therefore, the many-to-many relationship between investors and projects is resolved using the Investment entity, which records the investment amount, investment date, and investment status.

The Payment entity stores transaction information related to each investment. Every investment is associated with a single payment record containing the SSLCommerz transaction ID, payment method, payment status, and payment date. This establishes a one-to-one relationship between the Investment and Payment entities.

The ProgressUpdate entity records the development of farming projects by storing milestone information, descriptions, update dates, and optional progress images. Since a project may have multiple updates during cultivation, a one-to-many relationship exists between the Project and ProgressUpdate entities.

Overall, the ER Diagram provides a structured representation of the AgroInvest database by illustrating how farmers, investors, administrators, projects, investments, payments, progress updates, and profit reports are interconnected. This database design ensures efficient data management while supporting project verification, investment tracking, payment processing, project monitoring, and profit distribution.

4. Sequence Diagram

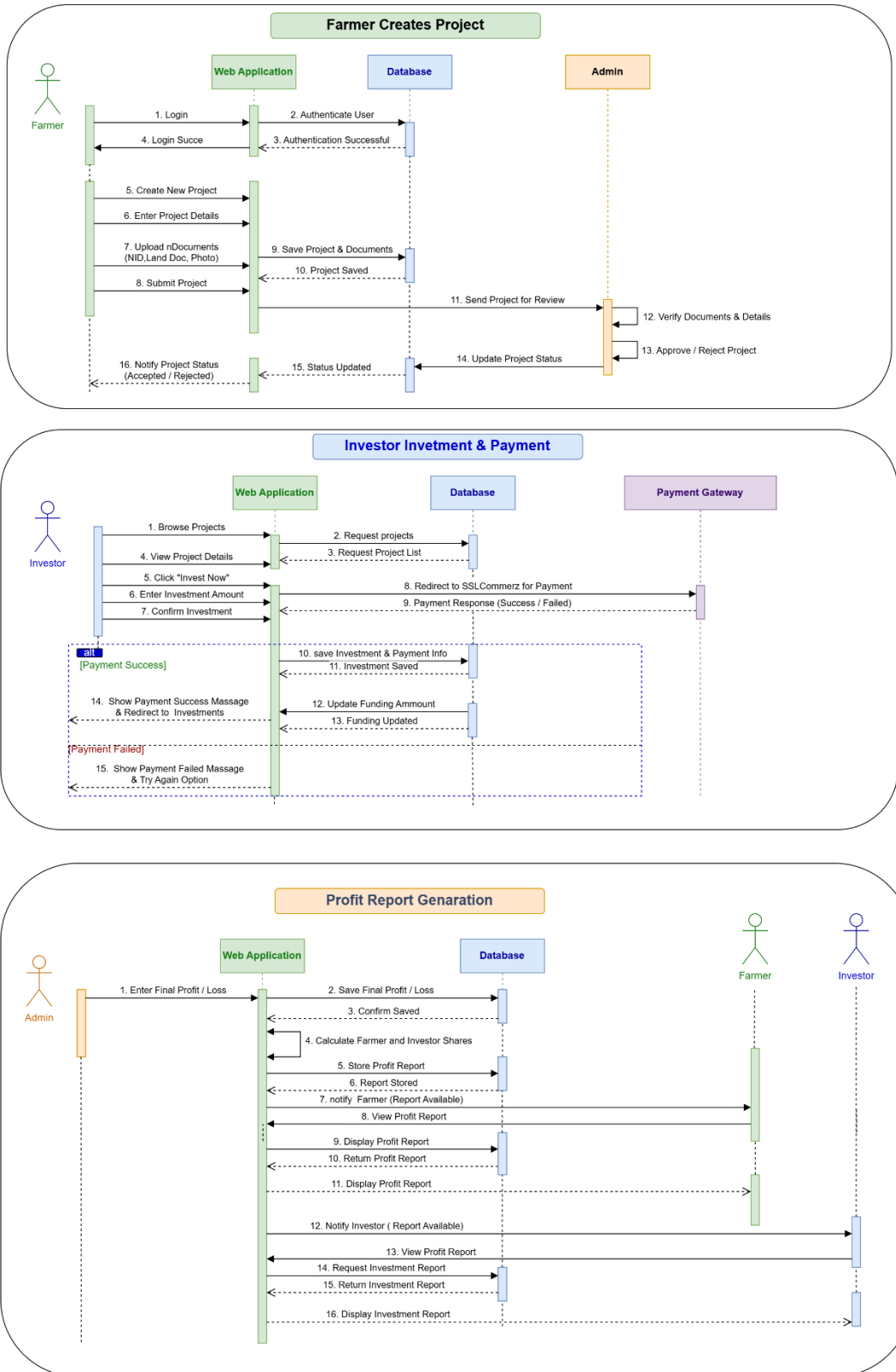


Figure 4: Sequence Diagram of AgroInvest System

The Sequence Diagram illustrates the chronological interaction between different system components involved in the AgroInvest platform. It demonstrates how messages are exchanged among the users, the web application, the database, the administrator, and the SSLCommerz payment gateway during three major business processes: Farmer Project Creation, Investor Investment and Payment, and Profit Report Generation.

The first sequence, Farmer Creates Project, describes the process of project submission. Initially, the farmer logs into the system and is authenticated by the web application through the database. After successful authentication, the farmer creates a new farming project by entering the required project details and uploading verification documents such as the National ID, land ownership documents, and land photographs. The web application stores the project information in the database and forwards the submitted project to the administrator for review. The administrator verifies the documents and either approves or rejects the project. Finally, the project status is updated in the database, and the farmer is notified of the approval or rejection result.

The second sequence, Investor Investment and Payment, represents the investment workflow. The investor browses the available projects, requests project information, and views the project details before selecting a project for investment. After entering the investment amount and confirming the investment, the web application redirects the investor to the integrated SSLCommerz Payment Gateway for secure payment processing. Based on the payment response, the system records the investment information in the database and updates the project's funding amount. If the payment is successful, the investor receives a payment success message and the investment is recorded in the "My Investments" section. If the payment fails, the investor is notified and provided with the option to retry the payment.

The third sequence, Profit Report Generation, illustrates the process of generating and publishing the final profit report after project completion. The administrator enters the final project profit or loss through the web application, which stores the information in the database. The web application then automatically calculates the profit shares for both the farmer and the investors based on the predefined profit-sharing policy. After the profit report is stored in the database, the system notifies both the farmer and the investors that the report is available. Finally, they can request and view their respective profit reports through the web application.

Overall, the Sequence Diagram provides a detailed view of the communication and message flow among different system components, ensuring that project submission, investment processing, payment handling, and profit distribution are performed in a structured and sequential manner.

5. Prototype Diagram

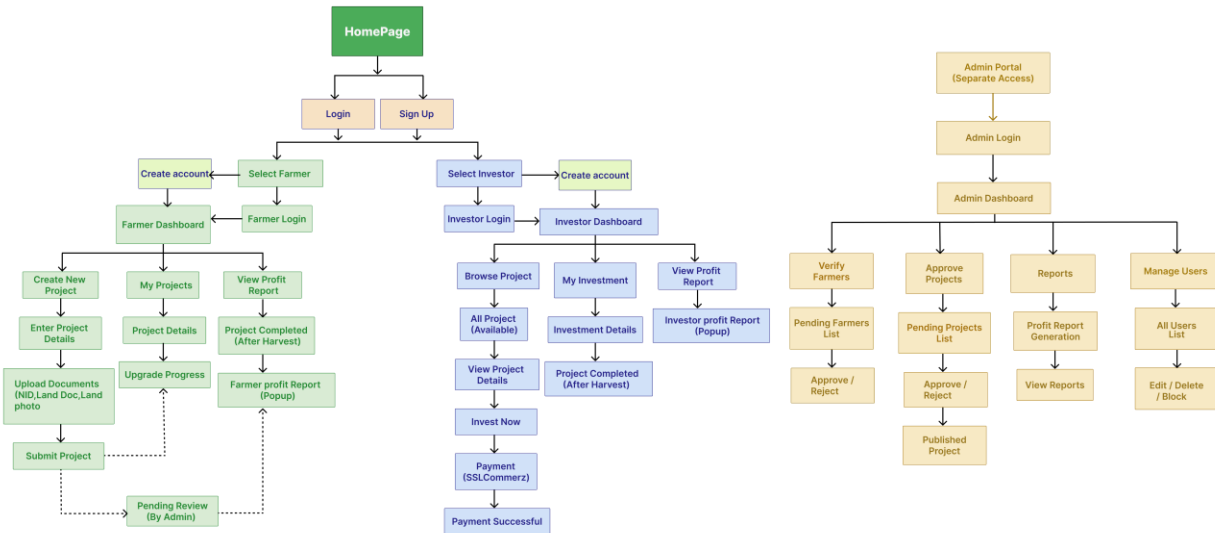


Figure 5: Prototype Diagram (Navigation Flow) of AgroInvest System

The Prototype Diagram illustrates the navigation flow of the AgroInvest system by demonstrating how users move between different screens of the application. It represents the user interface (UI) structure designed in Figma and provides a clear overview of the interactions among the Public Website, Farmer Module, Investor Module, and Admin Module. The diagram helps visualize the complete user journey before the actual system implementation.

The navigation begins from the Homepage, where users can either log in or create a new account. During registration, users select their role as either a Farmer or an Investor, after which they are redirected to their respective dashboards upon successful account creation or login.

The Farmer Module enables farmers to manage their agricultural projects. From the Farmer Dashboard, users can create a new project by entering project information, uploading verification documents such as the National ID, land ownership documents, and land photographs, and submitting the project for administrative review. After approval, farmers can access the My Projects section to view project details, update project progress, monitor project status, and view the final profit report after the farming project has been completed.

The Investor Module allows investors to browse available agricultural projects, view detailed project information, and invest securely through the integrated SSLCommerz Payment Gateway. After a successful payment, the investment is recorded in the My Investment section, where

investors can monitor their investment details, check project completion status, and access their individual profit report after the harvest has been completed.

The Admin Module is accessed through a separate and secure administrator portal. After logging in, administrators can verify farmer documents, approve or reject submitted projects, manage user accounts, monitor projects, generate profit reports, and maintain overall system operations. This separate access ensures better system security and restricts administrative functions from public users.

Overall, the Prototype Diagram provides a comprehensive visualization of the application's screen navigation and user interaction flow. It ensures that the interface is intuitive, logically organized, and consistent with the functional requirements defined in the Software Requirements Specification (SRS). The prototype also serves as a reference for implementing the final user interface and validating the overall user experience before system development.